

Structure-Based Prediction of NES Motifs using Graph Neural Networks

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Abstract

This project addresses the prediction of Nuclear Export Signals (NES) in protein structures using a graph-based representation of protein 3D data. We convert PDB files into residue-level graphs and train Graph Neural Networks (GNNs) to classify whether a protein contains NES motifs. Each residue is represented as a node enriched with features such as amino acid identity and spatial coordinates. Edges reflect spatial proximity, and the resulting data is suitable for modern geometric deep learning. Our pipeline enables direct use of structural information for biologically relevant predictions.

1 Introduction

Nuclear Export Signals (NES) are short amino acid motifs that enable proteins to be transported from the nucleus to the cytoplasm. Accurately predicting NES motifs is crucial for understanding protein localization and function, especially in the context of viral and cancer biology. Most previous methods rely on sequence-based predictors. In this work, we leverage 3D structural information via protein graphs to enable spatially-informed prediction using Graph Neural Networks (GNNs).

2 Data

2.1 Protein Structures

We use AlphaFold-predicted structures in PDB format. Our dataset contains 675 proteins, each labeled as either NES-positive or NES-negative based on curated annotations.

2.2 Preprocessing

Each PDB file is parsed using BioPython to extract C-alpha atoms representing amino acid residues. Labels are assigned based on external CSV metadata mapping each protein to its NES status.

3 Methods

3.1 Graph Construction Pipeline

We developed a complete pipeline to convert a protein’s 3D structure (in PDB format) into a graph-based representation suitable for geometric deep learning with PyTorch Geometric. The primary objective was to focus on nuclear export signal (NES) regions — specifically residues from chain B, which represent the NES motif — and their surrounding structural environment. To do this, we first filter the protein structure to retain only residues from chain B and any residues from other chains that lie within a 15 Ångström radius of

it, based on the position of their C-alpha atoms. This filtered structure can be saved as a new PDB file for inspection in PyMOL. We then build a graph where each node corresponds to a residue (specifically its CA atom), and each node is encoded using a 20-length one-hot vector for its amino acid type, with an additional binary feature indicating whether the residue belongs to the NES chain (chain B) or not. Edges are created between pairs of residues if their C-alpha atoms are closer than 8 Ångström, and the edge attributes store the actual Euclidean distances. The final output is a `torch_geometric.data.Data` object containing `x`, `edge_index`, `edge_attr`, and `pos`, ready to be passed into a GNN model. Additionally, we implemented a visualization function that plots the protein graph in 2D, highlighting NES residues in red and nearby residues in blue, along with the spatial proximity edges. This entire preprocessing framework enables us to isolate biologically meaningful substructures and represent them as spatial graphs for downstream tasks like NES classification.

3.2 Modeling Approach

We performed a thorough evaluation of several geometric GNN architectures, focusing on GVP-GNN and EGNN. Our selection criteria included data efficiency, geometric feature compatibility, and computational scalability. Below is a comparison table that highlights the critical aspects:

Feature	GVP-GNN	EGNN
Core Idea	Scalar/vector features	Embeddings + coordinates
Equivariance	SE(3) equivariant	Rotation, translation, reflection
Complexity	High (vector ops)	Moderate
Geometric Info Usage	Excellent for vectors	Excellent for distances
Data Efficiency	Lower, needs more data	Higher, great with ~300 samples
Scalability	Slower	Faster and scalable

Table 1: Comparison of GVP-GNN and EGNN

Given our use of C-alpha atoms and distance-based edge features without directional vectors, EGNN aligns perfectly with our graph representation. It efficiently handles small datasets and is computationally less intensive—ideal for hackathon time constraints. Thus, we selected EGNN for this project.

3.3 Baseline: Graph Convolutional Network (GCN)

Before selecting a more advanced architecture, we implemented a basic GCN as a baseline. GCNs are effective for graph-level tasks and help validate the data pipeline before incorporating geometric reasoning.

GCN works by updating each node’s features based on the average of its neighbors’ features. For NES classification, each residue (C-alpha atom) was treated as a node with one-hot amino acid encoding, and edges were added based on spatial proximity ($\leq 8\text{Å}$).

Strengths of GCN:

- Simple and efficient to implement.
- Fast training and a good pipeline validator.
- Reasonable performance using node features and graph connectivity.

Limitations:

- Ignores spatial geometry—no use of coordinates or distances.
- Lacks rotation or translation invariance.
- May fail to capture subtle structural motifs crucial to NES.

Conclusion: While GCN provided a useful performance baseline, it lacked geometric awareness, leading us to explore advanced architectures that incorporate 3D spatial information, such as EGNN and GVP-GNN.

4 Results

We initially visualized the full protein graphs including all residues and their spatial connections. **Figure 1** shows a dense representation of a protein graph before filtering. After cleaning the structure to focus on NES-relevant residues and nearby interacting regions, the resulting graph was significantly more interpretable. **Figure 2** illustrates the final NES-focused graph: NES residues are highlighted in red and nearby residues in blue, connected by spatially-defined edges.

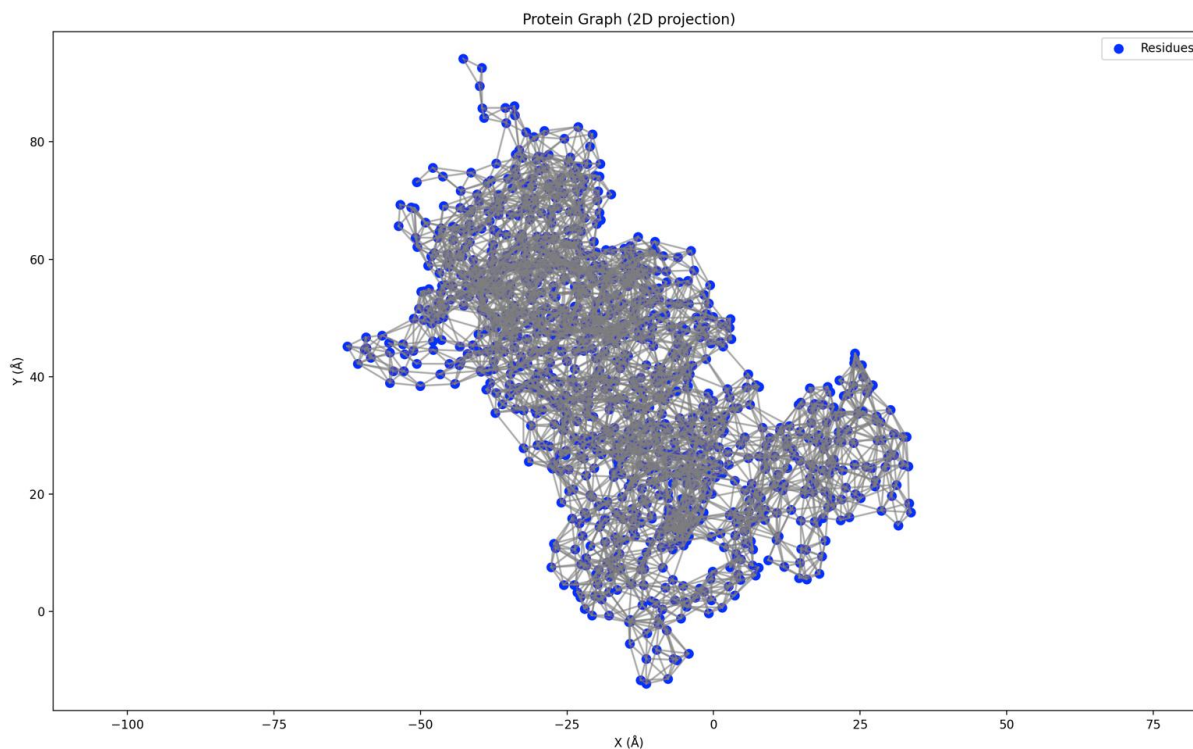


Figure 1: Initial dense protein graph including all residues.

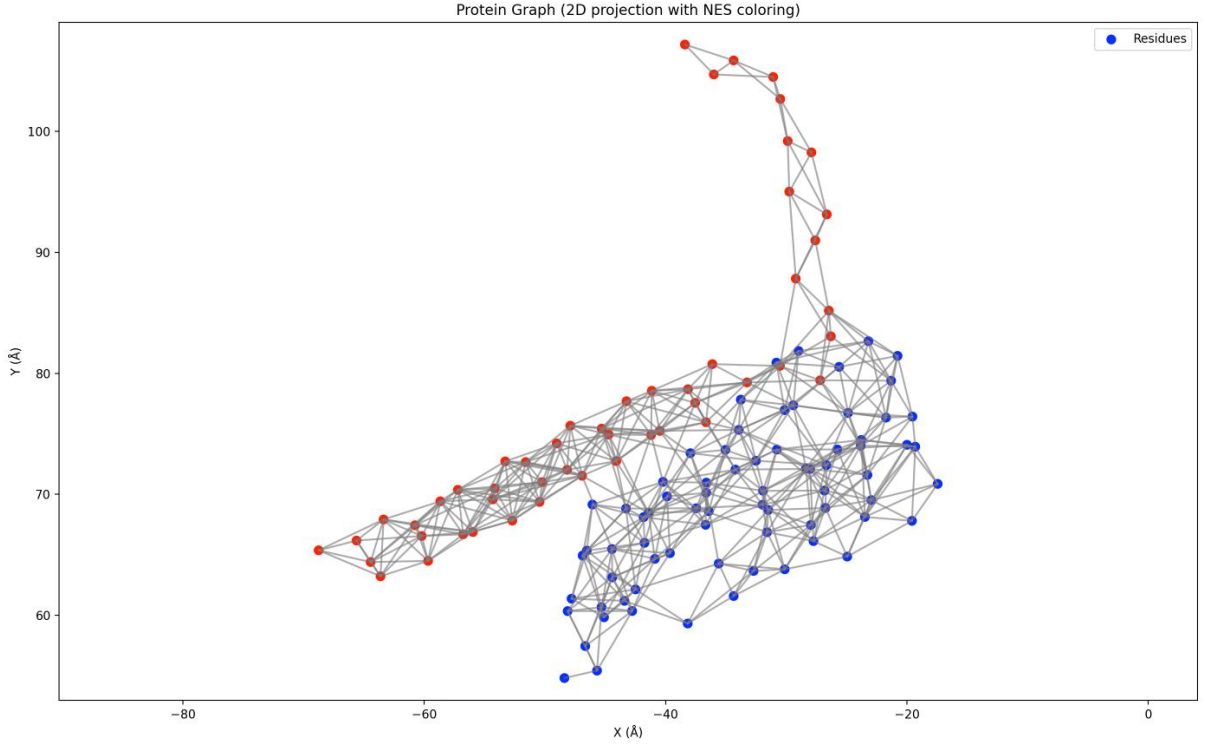


Figure 2: 2D projection of a protein graph showing residues from the NES chain (red) and nearby residues from other chains (blue), with edges representing spatial proximity.

5 Evaluation Metrics

To assess the classification performance of our EGNN model on the NES prediction task, we used two complementary evaluation methods: ROC curve analysis and score distribution visualization.

5.1 ROC Curve

The Receiver Operating Characteristic (ROC) curve shows the trade-off between the true positive rate (sensitivity) and false positive rate across all possible classification thresholds. A perfect classifier has a curve that hugs the top-left corner, while a random classifier follows the diagonal.

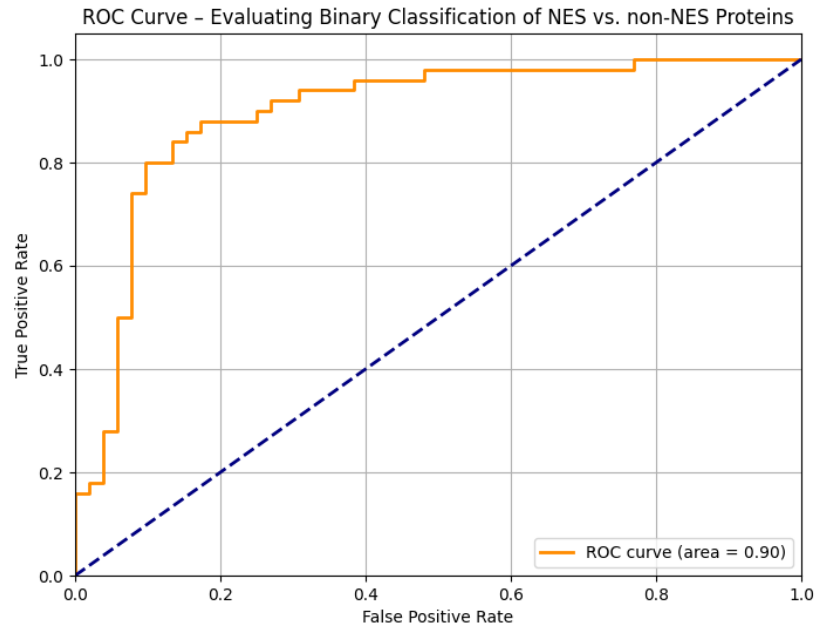


Figure 3: ROC Curve – Evaluating binary classification of NES vs. non-NES proteins. The area under the curve (AUC) is 0.90, indicating excellent discriminative ability.

5.2 Prediction Score Distribution

We also examined the distribution of the model's predicted probabilities (confidence scores) for NES-positive and NES-negative samples using a boxplot.

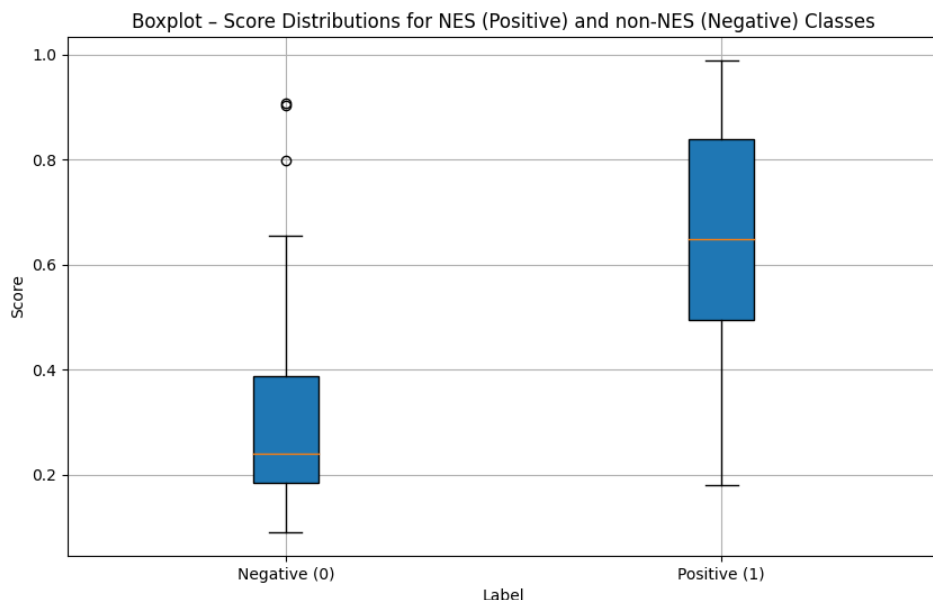


Figure 4: Boxplot showing predicted probability distributions. NES-positive samples tend to receive significantly higher scores than negatives, reflecting confident and well-separated predictions.

6 Discussion

Our graph-based approach allows us to use spatial information directly, potentially improving predictive accuracy over sequence-only methods. Preliminary results show promising distinction between NES and non-NES structures in graph form. EGNN’s efficiency and compatibility with CA-atom representations made it the optimal model for our constrained dataset size and structure.

7 Future Work

- Integrate ESM or pLDDT embeddings as node features.
- Incorporate interface-specific attention for NES pockets.
- Perform attention-weight visualization to identify critical residues.
- Explore additional architectures such as SE(3)-Transformer for comparison.

8 References

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